

Discovering Future Founders First: A Cross-Disciplinary Playbook for the Sourcing Layer of an AI-First VC

TL;DR

- **Founding is a rare, slow, multi-stage transition with real observable precursors.** Roughly a third of "nascent entrepreneurs" ever reach a profitable firm and about a third disengage within six years (Reynolds, PSED), while only ~37% of people who state an intention take meaningful action within a year (Kautonen et al. 2015; van Gelderen et al. 2018). The winning strategy is not to predict any one person, but to rank *populations* by founding yield and detect *state changes* (departures, repo creation, incorporation) early.
- **The best signals are people-level, not company-level, and they stack.** Employment departures from high-signal employers (spawning), stealth profile changes, new GitHub repos with product language, domain/entity registration, and hackathon/Product Hunt launches form a lead-time ladder from years-ahead to weeks-ahead. Commercial systems (EQT Motherbrain, SignalFire Beacon, Harmonic, Specter) already engage founders months before a round by fusing these; any single signal is too noisy, but two-plus combined yield high-confidence detection.
- **For a 24–72h hackathon, build a retrospective "we found them first" backtest.** Pick recently-funded startups, reconstruct their pre-funding trace timeline (first repo, domain, stealth title, incorporation), and show your detector would have flagged them months before Crunchbase/first press. Prioritize meritocratic, low-privilege signals (code, launches, filings) and measure time-to-detection, recall on a known-founder cohort, and outreach precision.

Key Findings

1. **Founding is predictable in the aggregate, not the individual.** Because the base rate of becoming a high-growth founder is tiny, intention data and population membership are more valuable than any single-person prediction. This is exactly Krueger & Carsrud's argument for intentions models: "in dealing with planned behaviours with a very low base rate of occurrence (such as starting a business), intentions offer significant insights."
2. **Prior employment is the strongest structural predictor.** Entrepreneurial "spawning" (Gompers, Lerner & Scharfstein 2005) and spinoff literature (Klepper) show that a small set of firms — VC-backed companies in Silicon Valley and Massachusetts — disproportionately produce founders. In 2014 a research analyst traced more than 92 public Bay Area tech companies (collectively worth more than \$2 trillion) to the founders and employees of Fairchild Semiconductor alone. Employer identity is a years-ahead signal.

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3. **Successful founders are middle-aged and industry-experienced**, not young dropouts. Azoulay et al. find the mean age at founding of the 1-in-1,000 fastest-growing ventures is 45.0, and prior industry experience strongly predicts success — reframing who to scan for. [SSRN](#)
4. **Sociological "liquidity events" spike founding in a population**: IPOs and acquisitions (especially by out-of-industry acquirers), and reductions in non-compete enforceability, measurably raise local founding rates (Stuart & Sorenson; Samila & Sorenson). These are calendarable, population-level triggers.
5. **Digital traces of pre-founding behavior are real and already commercialized**. Stealth LinkedIn titles, GitHub activity shifts, domain registrations, and incorporation filings are the working signal set of Harmonic, Specter, Evertrace, SignalFire, and EQT. People-level movement typically precedes any company-level footprint.
6. **Weak-signal detection has a mature methodological toolkit**: Kleinberg burst detection, change-point/anomaly detection, and operational event-forecasting systems (EMBERS for civil unrest) show how to forecast rare events from noisy public streams with lead time, and how to evaluate such systems honestly against base-rate baselines.
7. **Network position predicts future prominence**. Burt's structural-holes/brokerage results and bibliometric "rising star" prediction show that early centrality growth and bridging positions forecast success — transferable to GitHub co-contribution and co-authorship graphs.

Details

1. Nascent entrepreneurship & entrepreneurial intent

The gestation process and its base rates. The Panel Study of Entrepreneurial Dynamics (PSED I, ~1998–2000, $n \approx 830$ nascent entrepreneurs; PSED II, 2005 cohort with six annual interviews) is the canonical longitudinal dataset tracking people from the moment they begin "gestation" activities. It measures time in months from a "conception date" through transitions to an operating firm, quitting, or continued trying (Reynolds & Curtin). The headline base rates: about one-third of nascent entrepreneurs eventually develop a profitable firm, and about one-third disengage within the first six years, with a substantial remainder still "trying" for years (Reynolds, *Foundations and Trends in Entrepreneurship*, 2016 — "About a third will actually develop a profitable new firm," from a harmonized sample of ~2,500 nascent ventures across four countries; Reynolds & Curtin 2008). Short-horizon PSED transitions: roughly 30% had an operating business by 12 months, with large fractions still trying or giving up (Wagner, IZA DP 1293). [ICPSR + 4](#)

Intention → action gap. The Theory of Planned Behavior (Ajzen 1991), applied to entrepreneurship by Krueger & Carsrud (1993) and Krueger, Reilly & Carsrud (2000), holds that intention is the single best predictor of planned behavior, driven by attitude, subjective norms, and perceived behavioral control (PBC). Empirically, TPB constructs explain ~30–50% of

variance in intention, and intention plus PBC explains ~31% of variance in subsequent entrepreneurial action (Kautonen, van Gelderen & Fink 2015, *Entrepreneurship Theory and Practice*, n=969, Austria/Finland). But the gap is large: about 63% of people with entrepreneurial intentions had taken no or very little action a year later (van Gelderen et al. 2018, *Small Business Economics*), i.e., only ~37% converted to meaningful action within ~12 months. Moderators of conversion include self-efficacy (stronger for dissatisfied employees), fear of failure (weakens the intention→gestation link), and implementation intentions. (Taylor & Francis Online + 4)

Life-course and career predictors. Azoulay, Jones, Kim & Miranda ("Age and High-Growth Entrepreneurship," NBER 24489 / AER: Insights 2020) use US Census administrative data on 2.7M founders: mean founder age at founding is ~42 for firms with ≥1 employee, and 45.0 for the top 1-in-1,000 fastest-growing ventures; a 50-year-old founder is 1.8× more likely than a 30-year-old to achieve upper-tail growth; founders of patenting firms in Silicon Valley average 44.3. **Spawning:** Gompers, Lerner & Scharfstein (2005, *Journal of Finance*) show public firms in Silicon Valley/Massachusetts that were themselves once VC-backed are the most prolific spawners of new VC-backed ventures, favoring a "training and networks" explanation over the "frustrated innovator" one. **Spinoffs:** Klepper's work on Fairchild ("Fairchildren"), Detroit autos, and semiconductors shows the best firms spawn the most spinoffs, employees stay local, and spinoffs inherit parent capabilities; a 1986 SEMI genealogy chart traced 126 semiconductor companies directly to Fairchild. **Immigration:** the National Foundation for American Policy (Stuart Anderson, July 2022) found immigrants "have started more than half (319 of 582, or 55%) of America's startup companies valued at \$1 billion or more," ~64% of unicorns were founded by immigrants or their children, and "One-quarter (143 of 582, or 25%) of billion-dollar startup companies in the U.S. have a founder who first came to America as an international student."

(NBER + 3)

Sociological triggers that spike founding. Stuart & Sorenson (2003) show biotech IPOs and acquisitions near a region accelerate local founding rates — but acquisitions only when the acquirer is from outside the industry (cultural disruption releases talent). Non-compete enforceability strongly moderates these effects: Samila & Sorenson (2011) find stricter enforcement slows entrepreneurship, patenting, and employment growth; reduced enforceability (e.g., Massachusetts, Utah reforms) raised founding. Liquidity events create both capital and an alumni diaspora simultaneously.

2. Digital-trace signals of pre-founding behavior

Career-transition signals. Harmonic's own methodology names three stacked categories: **employment signals** (departure from a high-signal employer such as a top AI lab or growth-stage company at an inflection point; profile gaps with no stated next role), **network/hiring signals** (a founder recruiting a first technical co-founder or engineer), and **footprint signals** (a newly registered domain or fresh incorporation filing). "Any single category in isolation produces too many false positives; combined, they yield high-confidence detection." Stealth LinkedIn

titles ("building something new," title cleared with no new company) are explicitly flagged by Harmonic and Specter.

Turnover/attrition prediction (transferable methods). ML attrition models (SVM, XGBoost, random forests, survival analysis) use tenure, manager/re-org changes, pay, and network features; organizational-network research shows employees tied to leavers or on network peripheries are more likely to depart, and toxic culture is ~10× more predictive of attrition than pay (MIT Sloan / Sull). The transferable idea: *state-change* features (a manager change, a re-org, a colleague's departure) create an intervention/detection window — and a wave of departures from one firm is a spawning cluster.

Open-source signals of commercialization. VCs already scrape GitHub: Redpoint's Jordan Segall reported median GitHub stars of 2,850 at seed and 4,980 at Series A across 80 dev-tool companies, and confirmed "many VCs write internal scraping programs to identify fast growing GitHub projects." a16z's commercial-open-source framing emphasizes "project-community fit" measured by stars, contributors, commits, and PRs, and notes the project leader typically becomes the commercial CEO. The ROSS Index operationalizes this: it tracks vendor-led OSS startups under 10 years old below a funding threshold, using star growth rate as an early-momentum proxy (2025 leader Union Labs, 54.2× peak growth, 74.3K stars). **Caveat — signal gaming:** the peer-reviewed study "Six Million (Suspected) Fake Stars on GitHub" (Carnegie Mellon, NC State & Socket, ICSE 2026) analyzed 20TB of GitHub metadata (6.7 billion events, 326 million stars, July 2019–Dec 2024) with their tool StarScout and "identified 6.0 million fake stars and 18,617 repositories with fake star campaigns," generated by ~301,000 accounts; by mid-2024, 16.66% of all repositories with 50+ stars were involved in fake-star campaigns (near-zero before 2022), and stars sold for \$0.03–\$0.85 each. Any star-velocity detector needs fake-star filtering (account age, starring bursts, profile completeness). (Chatgpt + 2)

arXiv/patent signals. Patents act as credible signals that reduce founder-investor information frictions and causally improve financing, growth, and IPO/acquisition odds (Farre-Mensa et al. 2020; Hsu & Ziedonis). University spinout prediction: TTO age/experience and university-industry joint patents predict spin-off creation; but Roche & Boudou (Harvard) found startups relying heavily on academic research, and those where the academic inventor is also the founder, raise less and are less likely to be acquired — a nuance for scoring lab-to-startup transitions.

3. Weak-signal detection & forecasting methodology

Burst and change-point detection. Kleinberg's "Bursty and Hierarchical Structure in Streams" (2002) models an event stream with a two-state (or infinite-state) automaton that transitions between baseline and bursty states via a hidden Markov process, minimizing a cost that trades off state-transition penalties against fit; it outputs ranked bursts with weights and intervals. It is directly applicable to GitHub commit/star streams, arXiv keyword streams, and domain-registration streams, and has R (`bursts`) and Python (`burst_detection`) implementations.

Operational event forecasting — EMBERS. The IARPA-funded EMBERS system (Ramakrishnan et al., KDD 2014) forecasts civil unrest across 10 Latin American countries from open-source indicators (tweets, news, blogs, economic data), operational since Nov 2012, generating ~40–50 warnings/day, forecasting date, location (city resolution), event type, and population, with lead times of days and an audit trail. Design lessons for a founder-detection engine: a fusion/suppression engine combining multiple weak models via probabilistic soft logic; evaluation against base-rate baselines; and an auditable "why was this flagged" trail. EMBERS "beat the news" (forecast events before media reported them) — the exact analogue to "we found them before Crunchbase."

Lead-user theory (von Hippel 1986). Lead users sit at the leading edge of a trend and face needs months/years ahead of the market; they innovate for themselves and their innovations are disproportionately commercially attractive (Urban & von Hippel: 82% of a lead-user cluster had modified the product vs. 1% of non-lead users). Operationalization: first identify the trend, then find users at its leading edge — a template for scanning open-source/hackathon populations for people already building against a nascent need.

Rising-star / bibliometric prediction and network science. ML models (XGBoost, regression) predict future h-index with $R^2 \approx 0.93$ at 1 year and ~ 0.82 at 5 years; degree centrality correlates positively with h-index; diversity of collaborators predicts performance. Burt's structural-holes theory shows brokers spanning holes get disproportionate good ideas, promotions, and compensation (e.g., a 2004 study of 673 managers; low network constraint predicts early promotion). Preferential attachment and early centrality growth in collaboration networks (co-authorship, GitHub co-contribution) are candidate features for detecting a rising node *before* it becomes central.

4. Existing practice — how sourcing is actually automated

Proprietary VC platforms. EQT Ventures' **Motherbrain** (launched 2016) employs around 30 people across three divisions (Labs, Platform, Research), draws on roughly 40 data sources, and has scanned some 50 million companies on a regular basis (per research head Lele Cao); of ~50 investments since 2016, seven were directly sourced by Motherbrain, including AnyDesk and WarDucks — flagged *before the founders sought funding* — and Small Giant Games ($\rightarrow$ \$700M Zynga exit). It reportedly ingests headcount growth, web traffic, founder social activity, hiring composition, and news, and engages companies on average ~14 months before a funding round closes (a company-reported figure). **SignalFire's Beacon** (building since ~2013; costs >\$10M/yr to run) tracks, per recent reporting, ~650M people and ~80M organizations across 2M+ data sources — GitHub commit activity, app-store rankings, engineering-team growth by function, job-posting velocity, and domain registrations. **Harmonic** covers 35M+ companies and 195M+ people "from stealth through IPO," tracking talent below the executive layer so "the signal often appears at the individual level first," with stealth-tagging when someone clears a title or leaves without a new company. **Specter**, **Evertrace** (pre-formation focus: trade registries, patents, papers, grants, stealth GitHub, anonymous X accounts), **PredictLeads**, **SourceScrub**, and

Crunchbase round out the landscape. Andre Retterath's DDVC Landscape counted ~190 data-driven VC firms by 2024. A 2024 SSRN study found funds adopting ML sourcing invested more outside traditional hubs, and those deals were *more* likely to IPO/become unicorns.

Accelerator and university funnels. Y Combinator: ~5,000 companies funded since 2005, ~1% acceptance (Summer 2025 reportedly 0.6%), ~4.5% unicorn rate vs. ~2.5% for the broader seed market, and ~45% reach Series A vs. ~33% average. University production (Strebulaev data): Stanford leads by unicorn-founder count (147 founders / 113 unicorns), then MIT (117/89) and Harvard (86/76); on an odds-ratio (per-capita) basis, smaller programs like Utah (3.36×-3.72×), Washington (2.57×), and Vanderbilt (2.13×) over-index, and MIT is 1.93×. These per-capita rates are the right way to prioritize scanning by expected yield. Notably, one YC-founder analysis found founder background explains <4% of funding variance — a caution against over-weighting pedigree.

Recruiting-tech analogues. Passive-candidate signals used by technical recruiters — work anniversaries, company turmoil/layoffs, profile updates, tenure milestones — map directly onto founder-departure detection. SignalFire's Beacon Talent predicts, per company reporting, "how likely they are to switch jobs" for essentially the entire tech talent ecosystem; the same job-change propensity model is a founder-departure model when applied to talent at high-spawning firms.

5. Practical synthesis for a sourcing engine

Signal taxonomy by lead time (with detectability and precision/recall trade-offs):

Lead time	Signal	Data source	Detectability	Precision	Recall
Years ahead	Works at a high-spawning firm (VC-backed, SF/MA); recent PhD in a hot area; middle-aged with deep industry experience; immigrant/international student	LinkedIn, Census-style firmographics, university rosters, arXiv	Easy (population membership)	Very low	Very high
Years ahead	Sits in a brokerage/structural-hole position; rising centrality in co-authorship/co-contribution graph	GitHub graph, arXiv co-authorship	Medium	Low	Medium
Months ahead	Departure from high-signal employer with no stated next role; stealth LinkedIn/X title	LinkedIn, X	Medium	Low-medium	Medium
Months ahead	New GitHub repo with product language in README; sudden activity shift; license change; org creation; co-founder/first-engineer recruiting	GitHub API	Easy	Medium	Medium
Months ahead	Star-velocity burst on a new repo (fake-star-filtered)	GitHub API	Easy	Medium	Low-medium
Weeks ahead	Domain registration; incorporation filing (Delaware, Companies House); trademark/patent application	WHOIS, state/registry filings, USPTO	Easy-medium	High	Low
Weeks ahead	Hackathon win; Product Hunt / Show HN launch; accelerator cohort listing	Devpost, Product Hunt, HN, YC/accelerator directories	Easy	High	Low

The pattern mirrors Harmonic's rule: low-precision high-recall population signals (years-ahead) must be *combined* with high-precision low-recall event signals (weeks-ahead) to produce a high-confidence, actionable list. Two-plus independent signals crossing is the trigger.

Base-rate math and population prioritization. Because only ~a third of nascent entrepreneurs reach profitability and only ~37% of intenders act within a year, and high-growth founders are

far rarer still, expected yield per scanned person varies by orders of magnitude across populations. Rank scanning by (population founding propensity $\times$ detectability $\times$ strategic fit). Highest-yield populations: recent departers from a handful of high-spawning employers; recently-liquid alumni of a just-IPO'd or out-of-industry-acquired company; maintainers of fast-growing OSS projects; winners of top hackathons; recent PhDs/postdocs in a hot area with a first patent. GEM data anchor the funnel: $\sim 12\%$ of not-yet-active US adults intend to start within three years (rising to $\sim 21\%$ among 18-24-year-olds), but TEA (actually starting/running) sits near 19% and high-growth outcomes are a sliver of that. (Babson College)

How to evaluate a sourcing engine:

- **Time-to-detection vs. incumbents:** for a cohort of recently-funded companies, measure days between your first-flag date and (a) Crunchbase entry, (b) first press mention, (c) funding announcement. Target: consistently negative (earlier). This is EMBERS' "beating the news" metric.
- **Retrospective recall on known founders:** take a labeled set of founders who raised in the last 12-24 months; ask "could we have detected them earlier from their public trace?" Report recall and median lead time.
- **Outreach precision:** of surfaced-and-contacted people, what fraction were genuinely starting/about to start? Track precision@K for the ranked daily list.
- **Channel-yield feedback loops:** attribute eventual true-positives back to the originating signal/channel and re-weight; this is how EQT "cut 99% of deals early" without losing the winners.

Cold-start and fairness. Meritocratic, low-privilege signals: public code, launches, hackathon results, patents, and registry filings reflect *doing* rather than *pedigree/network*. Network-privilege proxies: "left a top lab," warm-intro graphs, and elite-school membership over-index on access and can entrench bias (and YC data show pedigree explains $< 4\%$ of funding variance). Mitigations: scan globally (Companies House, non-US registries, non-English GitHub/README), weight artifact-based signals over affiliation-based ones, and explicitly scan populations that don't emit standard signals (bootstrappers, non-VC ecosystems, builders who don't use LinkedIn). Evertrace/Harmonic's people-level, registry-and-GitHub-first approach is more meritocratic than press/Crunchbase-first databases.

Concrete 24-72h hackathon plan.

1. **Pick 3 public streams you can ingest fast:** GitHub (new repos + stars + README text via API), domain/WHOIS + one incorporation registry (Delaware or UK Companies House — both queryable), and Product Hunt/Hacker News "Show HN" launches. Optionally arXiv listings for a hot subfield.
2. **Build three detectors:** (a) a Kleinberg burst detector on repo star/commit streams (use the `bursts`/`burst_detection` libraries), (b) a keyword/LLM classifier that scores READMEs

and bios for "product language" and stealth phrasing, (c) an entity-join that cross-references a new domain/incorporation against a GitHub org or a departed-from-employer.

3. **Fuse and rank:** require ≥ 2 independent signals to fire; produce a daily ranked list with an audit trail (which signals, when) à la EMBERS.
4. **Demo "we found them first":** choose 5–10 startups that raised seed rounds in the last 6–12 months, reconstruct their pre-funding trace (first commit/repo, domain registration date, stealth title date, incorporation date, first launch) and overlay your detector's would-have-fired date vs. their Crunchbase/press date. A timeline showing your flag months earlier is the whole pitch.
5. **Report the metrics:** median lead time vs. Crunchbase, recall on the backtest cohort, and precision of the current day's live list.

Recommendations

1. **Start with people, not companies, and start with employers.** Build the "departure-from-high-spawning-firm" detector first: it is the highest-leverage years-ahead signal (Gompers/Lerner/Scharfstein; Klepper) and the one commercial leaders (Harmonic, SignalFire) lean on hardest. Benchmark: can you enumerate departers from a curated list of ~50 high-spawning firms weekly?
2. **Layer event signals to convert recall into precision.** Add GitHub-repo, domain, and incorporation detectors and require ≥ 2 signals to fire before surfacing. Threshold to change course: if precision@50 on your daily list is $< 20\%$, tighten the fusion rule; if recall on your backtest cohort is $< 50\%$, add a signal source rather than loosening thresholds.
3. **Instrument time-to-detection from day one.** Make "days earlier than Crunchbase/press" your north-star metric; it is measurable retrospectively today and is the credible demo. Aim to beat incumbents by a Motherbrain-style lead (its self-reported ~14 months) on at least a subset.
4. **Filter for signal gaming and privilege bias.** Implement fake-star filtering (account age, burst detection, profile completeness) given the ~6M documented fake stars and the fact that by mid-2024 ~1 in 6 repos with 50+ stars showed fake-star campaigns; weight artifact-based (code/launch/filing) over affiliation-based (elite lab/school) signals to widen the funnel and reduce bias, consistent with evidence that ML sourcing finds strong non-hub deals and that pedigree explains little funding variance.
5. **Prioritize populations by expected yield, and calendar the liquidity events.** Rank scanning by founding propensity $\times$ detectability; watch IPOs and out-of-industry acquisitions and non-compete law changes as population-level founding-rate spikes (Stuart & Sorenson; Samila & Sorenson) and pre-position outreach to those alumni diasporas.

Caveats

- **Base rates are brutal.** High-growth founding is extremely rare; most flagged people won't found, and most founders won't build high-growth firms. Precision will be low by construction — design for a ranked triage list feeding human relationship-building, not an automated verdict.
- **Signals are gamed and biased.** GitHub stars are purchasable at scale; LinkedIn/press signals over-index on privilege and access. Un-filtered, both degrade precision and entrench bias.
- **Vendor figures are self-reported.** The ~14-month Motherbrain lead, Beacon's data-source counts, and Harmonic's coverage claims come from the companies' own marketing/press and should be treated as directional, not audited; the ~14-month lead in particular could not be verified against an authoritative EQT primary source.
- **Academic base rates come from older/broader datasets.** PSED conversion rates, spawning studies (1986–1999 window), and Azoulay's age findings may not perfectly reflect the current AI-era, faster-formation environment; treat magnitudes as anchors, not precise present-day values.
- **A clean GitHub-user→founder base rate does not appear to exist** in the literature and should be estimated empirically from your own backtest; the PSED "∼1/3 profitable, ∼1/3 disengage" framing is the best-sourced conversion anchor available.
- **Correlation, not causation.** Structural-holes, spawning, and rising-star results are associational; using them as scanning priors is sound, but they don't guarantee individual outcomes.